BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021 • Paper Code: DIK-SSC-X-2021 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 12 | **Paper Code:** DIK-SSC-X-2021 (A)

Q1. Choose the correct option and fill in empty box, given against each statement:

Q1. In woody stems, the bark contains special pores for gaseous exchange called _____

(A) Stomata
(B) Lenticels
(C) Epidermis
(D) Cuticle

Q2. In kidney, process of pressure filtration for urine formation takes place in _____

(A) Glomerulus
(B) Bowman's capsule
(C) Loop of Henle
(D) Convoluted tubule

Q3. Which one of the following is made up of fatty material which insulates the zxon?

(A) Dendrites
(B) Cell body
(C) Myelin sheath
(D) Nodes

Q4. Hormones responsible for increasing the release of glucose from the liver to the blood is _____

(A) Insulin
(B) Glucagon
(C) Calcitonin
(D) Thyroxine

Q5. The bands of connective tissue which hold the bones at the joints in place are called _____

(A) Ligaments
(B) Tendons
(C) Cartilage
(D) Chondrocytes

Q6. In which type of asexual reproduction an outgrowth arises from the body of parent develop into offspring?

(A) Binary fission
(B) Multiple fission
(C) Spore formation
(D) Budding

Q7. In animals sex cells or gametes are produced in special organs called _____

(A) Gonads
(B) Antheridium
(C) Liver
(D) Kidneys

Q8. Who discovered thread like structures called chromosomes in a cell?

(A) Robert Hook
(B) Waldeyer
(C) Mendel
(D) Robert brown

Q9. _____ is a group of similar organisms that can breed with one another to produce fertile offspring.

(A) Genus
(B) Population
(C) Species
(D) Community

Q10. Earth's atmosphere contains almost _____ percentage of carbon dioxide.

(A) 4%
(B) 0.4%
(C) 40%
(D) 0.04%

Q11. In fermentation, the glucose molecule is first broken down into _____

(A) Pyruvic acid
(B) Lactic acid
(C) Citric acid
(D) Acetic acid

Q12. Medicinal drug used for the treatment of bacterial infection is_____

(A) Vaccines
(B) Antibiotics
(C) Sedatives
(D) Pain killers

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 21 | **Paper Code:** DIK-SSC-X-2021 (A)

Q2. Attempt any 8 questions of the following:

(i) Differentiate between respiration and breathing.

(ii) Give difference between transpiration and guttation.

(iii) Draw a labeled diagram of typical neuron.

(iv) Write a short note on thyroid gland.

(v) What are joints? Name different types of joints.

(vi) Write a short note on binary fission in amoeba.

(vii) What is meant by the term parthenogenesis?

(viii) Define gene and allele.

(ix) What is food chain? Give example.

(x) Why are vectors used in genetic engineering?

(xi) Write a short note on narcotics.

SECTION – C

Maximum Marks: 24

> Note: Attempt any 3 questions: Each carry 8 marks.

Q3.

(a) Describe structure and function of nephron.

(b) Write a note on spinal cord.

Q4.

(a) Describe axial skeleton in detail.

(b) Discuss spermatogenesis in animals.

Q5.

(a) Explain Mendel's law of segregation.

(b) Write a brief note on global warming.

Q6.

(a) What is alcoholic fermentation? Write its equation, diagrammatically.

(b) Write uses vaccines as a medicinal drug.